INCH-POUND

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DETAIL SPECIFICATION

CORD, AROMATIC POLYAMIDE, NONMELTING

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ASMC N/A FSC 4020

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1. SCOPE

1.1 Scope. This specification covers nonmelting aromatic polyamide cord.

1.2 Classification. The cord needs to be of the type specified (see Table I and 6.2).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in Sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in Sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

FED-STD-191	Textile Test Methods
MIL-STD-129	Military Marking for Shipment and Storage
MIL-STD-2073-1	Standard Practice for Military Packaging

(Copies of these documents are available online at https://quicksearch.dla.mil).

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D3951	Standard Practice for Commercial Packaging
ASTM-D5118/D5118M	Boxes, Fiberboard Shipping, Fabrication of
ASTM-D6251/D6251M	Natural Wood-Cleated Panelboard Shipping Boxes

(Copies of these documents are available from https://www.astm.org).

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/ASQ Z1.4	Sampling Procedures and Tables for Inspection by Attributes

(Copies of these documents are available from https://www.ansi.org).

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Government or supplier purchases. The requirements specified in 3.11 and 3.12 apply only to cord purchased directly by the Government. All other requirements apply to cord purchased by a supplier as a component for an end item and to cord purchased directly by the Government.

3.2 Standard sample. The cord shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.3 Materials. The yarn used in the manufacture of the cord shall be a nonmelting, aromatic polyamide. They yarn shall not carbonize at a temperature below 800 °F. The plied yarns shall be twisted only from the deniers specified in Table I.

3.4 Processing of yarns.

3.4.1 Stretching. The yarns used to manufacture the cord shall not be subjected to a stretching operation (see 4.2.2).

3.5 Bleaching. The yarn or the fabricated cord shall not be subjected to any type of bleaching process.

3.6 Construction and physical requirements. The fabricated cord shall conform to the applicable requirements specified in Table I and Table II and following paragraphs for the respective types when tested as specified in 4.2.1 and 4.2.6.

3.6.1 Lapping of core ends. Lapping of core ends is permissible providing the overlaps are more than 5-inches but do not exceed 10-inches in length (see 4.2.2). Splicing or knotting of core lines is not permissible (see 6.4). The minimum distance between a core end overlap or any adjacent core end shall be 500-feet. The approximate center of each overlap shall be marked by inserting a bright red thread through the sleeve. The free ends of the marker shall extend approximately 3-inches on either side of the sleeve. The removal of the marker shall be accomplished by a light pull on either end. When colored cord is specified, (see 3.8), the color of the marker shall be white.

NOTE: The core end marker shall not be removed until the cord is used in the manufactured end item.

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TABLE I. Construction.

	Type I (Coreless)	Type II		Type III	
		Core	Sleeve	Core	Sleeve
Yarns:					
Denier	200	200	200	200	200
Ply	4	1st – 7 Final – 3	4	1st – 7 Final – 3	4
Twist (turns/in.)					
Singles	7.0 to 9.55	Producers	7.0 to 9.5	Producers	7.0 to 9.5
Ply	5 to 7Z	1st 10 – 16S Final 6 – 8Z	5 to 7	1st 10 – 16S Final 6 – 8Z	5 to 7
Number of plied yarns			4	7 to 9	
Picks per inch	24 to 26		24 to 26		24 to 26
Number of carrier and ends per carrier	16/1		32/1		32/1

TABLE II. Physical requirements.

	Type I	Type II	Type III
Breaking strength, lbs., (min)	100	375	550
Elongation (%) (Min.)	20	20	20
Length/lb. (ft.) (Min.)	780	230	165

3.6.1.1 Protruding core ends. When any core end projects through the sleeve as a result of overlapping (see 3.6.1 and 6.4) the end(s) shall be cut flush with the surface of the sleeve with the cord in a relaxed condition.

CAUTION: Do not hot knife the end(s) and be extremely careful not to damage the sleeve.

3.6.2 Extractable matter (chloroform-soluble material). No material shall be added for the purpose of weighting the cord. The chloroform-soluble material of the cord shall not exceed 2.0 percent when tested as specified in 4.2.6.

3.7 Resistance to light and heat. The nylon cord shall lose no more than 15 percent of its original breaking strength after exposure to light and heat when tested as specified in 4.2.6.

3.8 Color. Unless otherwise specified (see 6.2), the color of the cord shall be natural. When colored cord is specified, the color shall be obtained by yarn dyeing, and the cord shall match an approved standard shade.

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3.8.1. Color matching. When dyed cord is specified, the shade of the cord shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500 Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800 Kelvin (see 6.3).

3.9 Type identification. Type II cord shall be identified by one international orange colored yarn braided into the sleeve (or cover).

3.10 Age. The yarn used to fabricate the cord shall be no more than two years old. The cord shall be no more than one year of age from the date of manufacture to the date of delivery to the first receiving point.

3.11 Put-up. Unless otherwise specified (see 6.2), the cord shall be put-up without knots in continuous lengths on spools (reels), as specified in the contract or order and in accordance with the lengths specified in Table III. Each spool (reel) of cord shall be so wound that each turn and layer is free from entanglement. A plus tolerance of 10 percent shall be allowed on the length of any spool (reel) specified. The ends of the cord shall be heat sealed to prevent fraying. Eighty percent of the total number of spools (reels) in the lot shall be of continuous length. The remaining twenty percent of the number of spools (reels) in the lot may contain pieced lengths (lapped on ends). These spools (reels) shall contain no more than two pieces with no piece less than 300-feet in length. When pieced lengths are contained on a spool (reels) the lengths of the pieces on the spool shall be marked on the identification ticket or label as specified in 3.12.

TABLE III. Length.

Type	Length on spool (reel), feet (minimum)
Type I	1500
Type II	2100 or 1200
Type III	2100 or 1200

3.12 Identification. Each spool (reel) of cord shall have a ticket (identification tag) or label attached to it for identification purposes. When labels are used, the label shall be attached in such a manner as to remain in place and be clearly legible until all cord has been removed. The ticket or label shall be legibly printed, stamped or typed with water insoluble ink. The ticket or label shall contain the following information:

Stock Number
Nomenclature
Specification Number
Length
Color
Contract number and date
Date of manufacture (month and year)
Length of pieces of cord per spool (when applicable)
Supplier’s name

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3.13 Workmanship. The fabricated cord shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in specification where such inspections are deemed necessary to assure supplies and services conform the prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with ANSI/ASQ Z1.4, except where otherwise indicated hereafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in the specification or applicable purchase documents. In addition, inspection shall be performed for the requirements in Table IV.

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TABLE IV. Component tests.

Characteristic	Paragraph reference	Test Method
Material	3.3	1/
Denier	3.3	1/
Melting point	3.3	1534 1/ 2/
Bleaching	3.5	1/
Age of yarn and cord	3.10	1/
Twist, single yarns	Table I	4052 1/ 2/ 3/
Twist, plied yarns	Table I	4052 1/ 2/ 3/

1/ Unless otherwise specified, a supplier's certificate of compliance is required and will be accepted for the requirements.

2/ Refers to test method of FED-STD-191.

3/ Twist of yarns: The number of specimens averaged for the result shall be five determinations on the core yarns and 10 determinations on the sleeve yarns, and the deadweight tension applied to the specimens shall be 60 ±2.0 grams. The 10 determinations of the sleeve yarns twist shall be made on specimens so selected that observations shall be made on five yarns braiding in one direction, and five yarns braiding in the opposite direction. Turns per inch of the single yarns (spin twist) shall be determined from the same specimen used for the determination of ply twist by cutting out all but one single yarn, leaving the plied ends still fastened in the clamps, and calculating on the basis of the elongated gage length which has resulted from the relief of the plied twist contraction.

4.2.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine that the requirements for processing of the core and sleeve yarns for shrinkage and stretching (see 3.4.1) and no more than 20 percent of the total number of spools (reels) in the lot contain pieced lengths of cord (see 3.12) have been adhered to in the manufacture of the cord. Unless otherwise specified, a supplier's certificate of compliance thereto shall be furnished and will be accepted for the requirements. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated non-conformance.

4.2.3 Examination of the end item for visual defects. The defects specified in Table V shall be counted, regardless of their proximity to each other, except where two or more defects represent a single local condition, in which case only the more serious defect shall be counted. The sample unit for the examination shall be one spool (reel). Ten percent of the length contained on each sample unit, but not less than 100-feet, shall be subjected to the visual examination. The lot size for this examination shall be expressed in units of spools (reels) each. The acceptable quality level shall be 1.5 major defects and 6.5 total defects (major and minor combined) per 100 units. Any critical defect found during sampling inspection shall be cause for rejection of the lot represented by the sample. The inspection level shall be level III. In regard to the critical defect class, inspection lot sizes of 150 or less shall be inspected 100 percent. For inspection lot sizes of 151 to 3,200, use a sample of 150 units. For inspection lot sizes greater than 3,200 use the sample size for level III, as indicated in ANSI/ASQ Z1.4. At its discretion, the Government may verify the result of the supplier's examination by 100 percent examination or by sampling

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inspection. Any critical defect found during sampling inspection shall be cause for rejection of the lot represented by the sample.

TABLE V. Visual examination - defects.

Examine	Defect	Classification		
		Critical	Major	Minor
Appearance and workmanship	Any cut or hole.	x		
	Abrasion, chafed area or distortion in the orientation of yarn.	x		
	Kink, or unevenly braided resulting in open place, break in continuity of braid or soft spot.	x		
	Float, broken or missing end or pick.	x		
	Any projecting core end.	x		
	Missing core end.	x		
	Type identification missing (when applicable).	x		
	Manufacturer's identification missing.		x	
Type	Other than specified.	x		
Put-up	Other than specified.		x	
Cleanness	Spot or stain clearly visible 1/.			x
	Objectionable odor.			x
Identification	Omitted		x	
	Information incorrect, illegible.			x
	Ticket or label not as specified.			x

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TABLE V. Visual examination - defects - Continued.

Examine	Defect	Classification		
		Critical	Major	Minor
Heavy, thick, or hard places, or uneven picks	Over 10 percent increase over maximum in picks per inch or over 2-inches of increased picks per inch.		x	
	10 percent increase over maximum in picks per inch for up to 2-inches.			x
Knots in plied yarn (per 10 linear yards)	Over 1 knot in sleeve yarns.		x	
	Any knot or splice in core.	x		
	1 knot in sleeve yarns.			x
Light place, pull down, step marks, or uneven picks	Over 10 percent decrease under minimum in picks per inch or over 2-inches of decreased picks per inch.		x	
Light place	10 percent decrease under minimum in pick per inch for up to 2-inches.			x

1/ At normal inspection distance (approximately 3-feet).

4.2.4 Examination for length and winding. The sample unit for this examination shall be one spool (reel). The inspection level shall be level S-3, and the acceptable quality level shall be 4.0 percent defective. For lots consisting of 500 or fewer units, the sample size shall be 10 and the acceptance number 1. The lot size shall be the number of spools (reels) in the lot. Defects shall be as listed in 4.2.4.1 and 4.2.4.2. The length on the spool (reel) (see Table III) shall be determined using the following formula:

W x L = Length per spool (reel) in feet.

When: W is the net weight of the spool (reel) in 0.1 pounds.

L is the lot average of the weight-length relationship (length/pound) as determined in 4.2.6.

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4.2.4.1 Defects with regard to length. Defects shall be considered to exist if any of the following are determined during inspection:

a. The length of cord on spool (reel) is less than specified of more than 10 percent in excess of the length specified.

b. Length of cord on spool (reel) more than 6-feet less than length marked on ticket or label.

4.2.4.2 Defects with regard to winding. Defects shall be considered to exist if any of the following are determined during inspection:

a. Improperly or not firmly wound resulting in kinking, knotting, entangling or slippage during unwinding or otherwise affecting free unhampered unwinding of cord.

b. Put-up not as specified.

c. Any end not heat sealed.

d. Cord not in a continuous length.

e. Knot or otherwise joining of ends to make a continuous length.

f. Any core end lap less than 500-feet apart (when applicable).

g. Any spool (reel) found to contain more than two pieces of cord.

h. Any piece of cord less than 300-feet in length.

i. The lengths of the pieces not marked on the identification ticket or label.

4.2.5 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging and packing requirements of Section 5 of this specification are complied with. The examination shall be in accordance with provisions of MIL-STD-2073-1, except that the inspection level shall be ANSI/ASQ Z1.4 S-2 and the AQL shall be 2.5 defects per 100 units.

4.2.6 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable and as listed in Table VI shall be followed. When the data in the "Number of determinations" and "Results reported as" columns are not specified in the table, they shall be as required by the referenced test method. The physical and chemical values specified in Section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. Sampling for testing shall be in accordance with ANSI/ASQ Z1.4, inspection level S-3. The sample unit for testing shall be one spool (reel) of cord. The lot size shall be expressed in units of spools (reels) of cord each. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more units fail to meet any requirement. The sample unit for testing the length of the overlap of core end shall be one lap marked spool (reel) of cord. One determination shall be made per sample unit. The sample size (number of sample units shall be three lap marked spools (reels) regardless of the lot size. The lot shall be unacceptable if one or more units fail to meet the overlap length requirement specified in 3.6.1. Tests to determine compliance with specification

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requirements, including quantity of delivery, may be made under prevailing atmospheric condition. In cases of dispute, tests shall be made upon material which has reached equilibrium under standard conditions as defined in ANSI/ASQ Z1.4.

TABLE VI. Test methods.

Characteristic	Requirement reference	Test method	No. of determinations per individual sample unit	Results reported as
Number of carriers	Table I	Visual	3	Avg of 3 determinations to nearest whole no.
Number of ends per carrier	Table I	Visual	3	Avg of 3 determinations to nearest whole no.
Ply	Table I	Visual	3	Avg of 3 determinations to nearest whole no.
Length per pound	Table II	4.2.6.2	3	Avg of 3 determinations to nearest 0.1 foot.
Breaking strength	Table II	4102	5	-
Elongation	Table II	4102	5	-
Extractable matter (chloroform soluble material)	3.6.2	2611	-	-
Resistance to light	3.7	4.2.6.3	-	-
Resistance to heat	3.7	4.2.6.4	-	-
Length of overlap of core ends	3.6.1	4.2.6.5	1	Pass or fail

4.2.6.1 Determination of picks per inch. Picks per inch shall be counted visually with the cord so positioned that the repeat of the stitch pattern creates a straight line parallel to the axis of the cord.

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4.2.6.2 Determination of length per pound. A specimen of cord shall be put under a tension load equal to 1 percent of the minimum specified breaking strength for the applicable type. With the specimen under load a minimum length of 10-feet shall be accurately measured and marked on the specimen to the nearest 0.25-inch. After marking, the tension load shall be removed, and the length measured shall be cut out of the test specimen and accurately weighed to within plus or minus 1.0 percent of its total weight. The length per pound shall be calculated utilizing the length measurement obtained under load. Three determinations shall be made per sample unit and the results averaged to the nearest 0.1-foot.

4.2.6.3 Determination of resistance to light. Five test specimens shall be taken from each sample unit. The test specimens shall be exposed in the accelerated weathering unit as specified in Method 5804 of FED-STD-191. The specimens shall be placed side by side on the rack in such a manner that only the center portion of each specimen is exposed. Unbacked specimens shall be exposed either mounted in stainless steel frames or as a large piece suspended from the outside rim of the exposure rack. Corex D filters and sunshine carbons shall be used. The exposure time shall be 50 hours. The spray heads shall be shut off during the entire exposure period and the relative humidity in the test chamber during the exposure shall be 55 ±5 percent. At the end of the exposure period, the specimens shall be conditioned at 70 ±2 °F (21 ±1 °C) and 65 ±2 percent relative humidity for 24 hours. The specimens shall then be tested for breaking strength in accordance with Method 4102 of FED-STD-191, and the percent of breaking strength (B.S.) lost shall be calculated as follows:

$$\text{Percent of B.S. lost} = \frac{\text{Original B.S.} - \text{B.S. after expoxure}}{\text{Original B.S.}} \text{ X } 100$$

One determination shall be made per individual sample unit.

4.2.6.4 Determination of resistance to heat. Five test specimens shall be taken from each sample unit. The test specimens shall be suspended in a circulating air oven at a temperature of 180 ±3 °C (356 ±5 °F) for 1 hour. After removal from the oven, the specimen shall be conditioned at 70 ±2 °F and 65 ±2 percent relative humidity for 24 hours. The specimens shall then be tested for breaking strength in accordance with Method 4102 of FED-STD-191, and the percent breaking strength lost shall be calculated as specified in 4.2.6.3. One determination shall be made per individual sample unit.

4.2.6.5 Determination of the length of overlap of core ends. A length of cord (approximately 4-feet) with the lap marker thread located at the center of the specimen shall be cut out of the spool (reel). Remove the lap marker thread and carefully remove the sleeve of the cord in such a manner as not to disrupt the core line ends. After removal of the sleeve the overlap length of the core ends shall be measured, and the results reported as pass or fail.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. Spools (reels) of cord, put up as specified shall be packaged in accordance with the applicable requirements of MIL-STD-2073-1.

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5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C. Spools (reels) of cord shall be packed in accordance with the applicable requirements of ASTM D3951, and the shipping container shall conform to ASTM-D5118/D5118M, style RSC, Grade V2s or ASTM-D6251/D6251M Class 2, Type 2, Styles A or J, as specified (see 6.2).

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The cord covered by this specification is intended for use as personnel parachute suspension lines and is also used with various equipage items.

6.2 Ordering data.

a. Title, number and date of this specification.

b. Type required (see 1.2 and Table I).

c. Color when other than natural (see 3.8).

d. Put-up if other than specified (see 3.12).

e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

f. Type of shipping container desired for level A packing (see 5.2.1).

6.3 Standard sample. For access to the standard sample, address the procuring office issuing the invitation for bids.

6.4 Splicing as used in this specification is defined as the joining of two strands or core ends by process of interweaving or mechanical joint. Lapping of core ends as used in this specification is defined as the insertion of an incoming core line running parallel to a running-out core line without splicing, i.e. to lie upon and extend between two given points.

6.5 Subject term (key word) listing.

Denier
Extractable matter
Spool
Winding
Yarn

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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